

Lecture Syllabus

Module Code	Semester	ECTS	SWS	Lecture	Tutorial	Lab
MECH-B-4-MLDS-MLDS1-ILV	4	5	4	0	0	0
Course Name	Data Science I					
Lecturer	D. T. McGuiness, Ph.D (Daniel.McGuiness@mci.edu) (4A-434c)					
Study Programme	Mechatronic Design Innovation					
Official Name	Machine Learning and Data Science 1				Lingo	English
Lecture Prerequisites	The student should be comfortable with working with the Python and have a good background with mathematics.					
Course Objectives	The goal of this lecture is to give you an introduction to the ever-growing industry of machine learning and its other side of the coin, data science.					
Primary Course Content	Lecture Homepage on GitHub WebBook					
Secondary Course Content(s)	<i>Probability: A Graduate Course</i> by A. Gut , <i>Probability and Statistics for Engineers & Scientists</i> by R. E. Walpole, et. al , <i>Differential Equations with Applications and Historical Notes (3rd Edition)</i> by G. F. Simmons , <i>Applied Statistics and Probability for Engineers (3rd Edition)</i> by D. C. Montgomery ,					
Homework(s) and Project(s)	Personal Assignment A (50) Personal Assignment B (50)					
Assessment Criteria	Assignment Type			Effect	Count	
	Personal Assignment A			50	1	
	Personal Assignment B			50	1	

Lecture Structure

Order	Topic	Units	Self Study
1	Theory of Probability	8	16
	<i>Introduction · Probability · Permutations and Combinations · Random Variables and Probability Distributions · Mean and Variance of a Distribution · Binomial, Poisson, and Hyper-geometric Distributions · Normal · Distribution of Several Random Variables</i>		
2	Statistical Methods	8	16
	<i>Introduction · Point Estimation of Parameters · Confidence Intervals · Testing of Hypotheses and Making Decisions · Quality Control · Acceptance Sampling · Goodness of Fit · Non-parametric Tests · Regression and Correlation · Bayesian Statistics</i>		
3	Regression Models	12	24
	<i>Simple Linear Regression and Correlation · Multiple Linear and Nonlinear Regression · Study of Residuals and Violation of Assumptions</i>		
4	Single Factor Experiments	4	8
	<i>Analysis-of-Variance Technique · One-Way Analysis of Variance · Tests for the Equality of Several Variances · Single-Degree-of-Freedom Comparisons · Multiple Comparisons · Comparing a Set of Treatments in Blocks · Randomized Complete Block Designs · Graphical Methods and Model Checking · Data Transformations in Analysis of Variance · Random Effects Models · Case Study</i>		
5	Factorial Experiments	4	8
	<i>Interaction in the Two-Factor Experiment · Two-Factor Analysis of Variance · 2 over k Factor Experiments · The Orthogonal Design</i>		
6	Nonparametric Statistics	4	8
	<i>Nonparametric Tests · Signed-Rank Test · Wilcoxon Rank-Sum Test · Kruskal-Wallis Test · Runs Test · Tolerance Limits · Rank Correlation Coefficient</i>		
7	Bayesian Statistics	4	8
	<i>Initial Concepts · Bayesian Inferences · Bayes Estimates Using Decision Theory Framework</i>		
8	The Essentials of Machine Learning	8	16
	<i>Defining Machine Learning · Types of Machine Learning Systems</i>		
9	A Machine Learning Project	8	16
	<i>Introduction · Working with Real Data · Downloading the Data · Creating a Test Set · Visualising Data · Preparing the Data</i>		
10	Sum	60	120

- Any major announcements will be made on SAKAI regarding any possible date/content/structural changes for the assignment(s), exam(s).
- Any lecture material will be posted at the lectures corresponding GitHub home-page. The link will be present on the lectures SAKAI homepage.
- If there are any questions regarding course content/exams/assignments please do not refrain from contacting me (Daniel.McGuiness@mci.edu).